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Underwater propulsion device

Abstract

An underwater propulsion device is disclosed comprising two sleeves for fitting around each of a user's lower legs, with each sleeve mounting a propulsion unit, and the sleeves being connectable by a bar between them during underwater operation of the device by the user.

Inventors: Robinson; Brandon (West Palm Beach, FL), Robinson; Lowell Kim (Sanford, FL), Barber; Marc (Deltona, FL)

Applicant: Robinson; Brandon (West Palm Beach, FL); Robinson; Lowell Kim (Sanford, FL); Barber; Marc (Deltona, FL)

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Assignee: Robinson; Brandon (West Palm Beach, FL); Robinson; Lowell Kim (Sanford, FL); Barber; Marc (Deltona, FL)

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Primary Examiner: Olson; Lars A

Attorney, Agent or Firm: Foley & Lardner LLP

Background/Summary

CROSS-REFERENCE TO RELATED PATENT APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 18/473,968, filed Sep. 25, 2023, which is a continuation of U.S. patent application Ser. No. 17/826,075, filed May 26, 2022, now U.S. Pat. No. 11,926,402, granted Mar. 12, 2024, which is a continuation of U.S. patent application Ser. No. 16/936,134, filed Jul. 22, 2020, now U.S. Pat. No. 11,345,453, granted May 31, 2022, which claims priority from U.S. Provisional Application No. 62/922,011, filed Jul. 22, 2019, and is a continuation-in-part of U.S. patent application Ser. No. 16/785,361, filed Feb. 7, 2020, now U.S. Pat. No. 11,173,345, granted Nov. 16, 2021, which is a continuation of U.S. patent application Ser. No. 16/115,392, filed Aug. 28, 2018, now U.S. Pat. No. 10,576,332, granted Mar. 3, 2020, which is a continuation of U.S. patent application Ser. No. 15/916,235, filed Mar. 8, 2018, now U.S. Pat. No. 10,071,289, granted Sep. 11, 2018, and which in turn claims priority from U.S. Provisional Application No. 62/469,129, filed Mar. 9, 2017, and U.S. Provisional Application No. 62/590,238, filed Nov. 22, 2017. The contents of all aforementioned applications are incorporated herein by reference in their entireties.

FIELD OF INVENTION

(1) The present invention relates to providing accessories for a battery powered propeller driven foot-mounted board for a swimmer or diver.

BACKGROUND OF THE INVENTION

(2) Known in the art are underwater snorkel or diver hand-operated propulsion devices. For example, the Sea Doo® RS series devices are battery powered using lithium ion lightweight batteries. The handlebar controls are used to hold the device in front of the diver. The unit has a neutral buoyancy. Squeezing two triggers with one's hands powers the unit, and releasing the triggers stops the power to the propeller. Apart from requiring hand operation, such devices tend to have minimal thrust. As used herein, pre-existing hand-held thrust units will be referred to as hand-held propulsion units or generically as "sea scooters."

(3) There is a need in the art to devise a system for adapting existing hand-held propulsion units to be capable of being mounted to a user's back, chest, or feet.

(4) Beyond such an adaptor system, there is a need for a stand-alone device unlike any in the prior art hand-held propulsion units that is specifically designed to be foot-mounted, to be activated by the user's feet, and to allow substantial thrust underwater.

(5) There is also a need for accessory equipment to facilitate use of the novel foot-mounted underwater propulsion system described herein.

SUMMARY OF THE INVENTION

(6) One aspect of the present invention is to provide a kit that clamps onto a hand-held propulsion device and enables mounting to a user's chest, back, or feet.

- (7) Another aspect of the present invention is to provide a novel device specially designed to be foot-mounted. In one embodiment, the device may take the form of an underwater foot board with an integral battery and motor with one or more propellers. Another embodiment of the inventive foot-mounted propulsion unit provides for a swivel foot mount to control a cable or an electronic switch that controls the speed of the motor.
- (8) Another aspect of the present invention is to provide accessory equipment for use with the novel foot-mounted underwater propulsion technology described herein.
- (9) Other aspects of this invention will appear from the following description and appended claims, reference being made to the accompanying drawings forming a part of this specification wherein like reference characters designate corresponding parts in the several views.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a front elevation view of a strap on foot board and a rear mounted board.
- (2) FIG. 2 is a front elevation view of a clip on foot board and a rear mounted foot board.
- (3) FIG. 3 is a front elevation view of a handle mounted foot board.
- (4) FIG. 4 is a front elevation view of a top mounted foot board.
- (5) FIG. 5 is a front elevation view of a dual scooter swivel foot board.
- (6) FIG. 6 is a front cross-sectional view of an integral battery powered foot board.
- (7) FIG. 7 is a top plan view of the FIG. 6 embodiment.
- (8) FIG. 8 is a front cross-sectional view of a dual motor integral battery powered foot board.
- (9) FIG. 9 is a top plan view of the FIG. 8 embodiment.
- (10) FIG. 10 is a front perspective view of embodiment of the device.
- (11) FIG. 11 is a side perspective view of a sea scooter fitted with a cable driven throttle button lever.
- (12) FIG. 12 is a perspective view of the throttle button lever assembly mounted to a sea scooter hand grip.
- (13) FIG. 13A is a side view of the throttle button lever assembly.
- (14) FIG. 13B is a perspective view of the throttle button lever assembly.
- (15) FIG. 13C is a side view of the throttle button lever assembly.
- (16) FIG. 13D is a side cross-sectional view of the throttle button lever assembly.
- (17) FIG. 13E is a top view of the throttle button lever assembly.
- (18) FIG. 14 is an exploded view of the throttle button lever assembly.
- (19) FIG. 15 is a front perspective view of a foot controlled foot board.
- (20) FIG. 16 is a bottom perspective view of the foot controlled foot board.
- (21) FIG. 17 is a bottom plan view of the foot controlled foot board.
- (22) FIG. 18 is a bottom perspective view of an embodiment of the device.
- (23) FIG. 19 is a bottom plan view of an embodiment of the device.
- (24) FIG. 20 is a top plan view of an embodiment of the device.
- (25) FIG. 21 is a side view of an embodiment of the device.
- (26) FIG. 22 is a top perspective view of the embodiment of the device.
- (27) FIG. 23 is a bottom perspective view of the embodiment of the device.
- (28) FIG. 24 is an exploded view of the embodiment of the device.
- (29) FIG. 25 is a front perspective view of the embodiment of the device mounted to a sea scooter.
- (30) FIG. 26 is a top plan view of a back mounted sea scooter.
- (31) FIG. 27 is a side perspective view of an L bracket back embodiment.
- (32) FIG. 28 is side elevation view of an L bracket chest embodiment.
- (33) FIG. 29 is a front view of a dual L bracket foot board.

- (34) FIG. 30 is a front view of a dual L bracket foot board.
- (35) FIG. 31 is a front elevation view of a quick disconnect boot embodiment.
- (36) FIG. 32 is a front cross-sectional view of a quick disconnect boot locked into place.
- (37) FIG. 33 is a bottom plan view of a foot pedal magnet based speed control embodiment.
- (38) FIG. 34 is a top perspective view of the FIG. 33 embodiment.
- (39) FIG. 35 is an exploded view of the FIG. 33 embodiment.
- (40) FIG. 36 is a top plan view of a foot pedal.
- (41) FIG. 37 is a top plan view of the foot board and kill switch.
- (42) FIG. 38 is a diagram of the subsystems of the electronic control system.
- (43) FIG. 39 is a flowchart of an embodiment of the control logic.
- (44) FIG. 40 is a top plan view of a sample hand control wireless embodiment controller.
- (45) FIG. 41A is a front elevation view of another embodiment of the device.
- (46) FIG. 41B is another front elevation view of the embodiment in FIG. 41A.
- (47) FIG. 42A is a front view of another embodiment of the device.
- (48) FIG. 42B is another front view of the embodiment in FIG. 42A.
- (49) FIG. 42C is a front elevation view of the embodiment in FIG. 42A.
- (50) FIG. 43 is a front elevation view of another embodiment of the device.
- (51) FIG. 44 is a front elevation view of another embodiment of the device.
- (52) FIG. 45 is a side cross-sectional view of another embodiment of the device.
- (53) FIG. 46 is a front elevation view of another embodiment of the device.
- (54) FIG. 47A is a top view of another embodiment of the device.
- (55) FIG. 47B is a side view of the embodiment shown in FIG. 47A.
- (56) FIG. 48A is a top view of another embodiment of the device.
- (57) FIG. 48B is a side view of the embodiment shown in FIG. 47A.
- (58) FIG. 49A is a front elevation zoom view of an element of an embodiment of the device.
- (59) FIG. 49B is a view of an element of the device attached to a human leg.
- (60) FIG. 49C is a side view of an element of an embodiment of the device.
- (61) FIG. 50 is a side view of an element of an embodiment of the device in isolation, and also shown attached to a user.
- (62) FIG. 51 is a side view of an element of an embodiment of the device.
- (63) FIG. 52 is a side perspective view of an element of an embodiment of the device.
- (64) FIG. 53A is a perspective view of an element of an embodiment of the device.
- (65) FIG. 53B is a perspective view of an element of an embodiment of the device.
- (66) FIG. 54A is a perspective view of an element of an embodiment of the device.
- (67) FIG. 54B is a perspective view of an element of an embodiment of the device.
- (68) FIG. 55A is a side view of two element of an embodiment of the device.
- (69) FIG. 55B is a side view of one of the elements of FIG. 55A shown in different stages of operation.
- (70) FIG. 55C is a side view of elements of an embodiment of the device.
- (71) FIG. 55D is side view of one of the elements shown in FIG. 55C.
- (72) FIG. 55E is a side view of the element from FIG. 55D shown at a different stage of operation.
- (73) FIG. 56A is a side view of an element of an embodiment of the device.
- (74) FIG. 56B is a side view of one of the elements shown in FIG. 56A.
- (75) FIG. 57A is a side view of an element of an embodiment of the device.
- (76) FIG. 57B is a side view of an embodiment of the device being worn by a user.
- (77) FIG. 58 is a rear perspective view of elements of an embodiment of the device.
- (78) FIG. 59A is a side view of elements of an embodiment of the device.
- (79) FIG. 59B is a side view of an element shown in FIG. 59A.
- (80) FIG. 60 is a side perspective view of elements of an embodiment of the device.
- (81) FIG. 61 is a side view of elements of an embodiment of the device.

- (82) FIG. **62** is a top view of elements of an embodiment of the device.
- (83) FIG. **63** is a planar view of an element of an embodiment of the device.
- (84) FIG. **64** is a side view of elements of an embodiment of the device.
- (85) FIG. **65** is a side view of elements of an embodiment of the device.
- (86) FIG. **66** is a side view of elements of an embodiment of the device.
- (87) FIG. **67** is a side view of elements shown in FIG. **66**.
- (88) FIG. **68** is a side view of elements of an embodiment of the device.
- (89) FIG. **69A** is a side view of elements of an embodiment of the device.
- (90) FIG. **69B** is a side view of elements of an embodiment of the device.
- (91) FIG. **70** is a side view of elements of an embodiment of the device.
- (92) FIG. **71** is a front perspective view of an embodiment of the device.
- (93) FIG. **72** is a front perspective view of the embodiment shown in FIG. **71**.
- (94) FIG. **73** is a detail view of the embodiment shown in FIG. **72**.
- (95) FIG. **74** contains a top and front view of the embodiment shown in FIG. **71**.
- (96) FIG. **75** is a front perspective view of the embodiment shown in FIG. **71**.
- (97) FIG. **76** is a diagram of the electronic and computer control elements of an embodiment of the device.
- (98) Before explaining the disclosed embodiment of the present invention in detail, it is to be understood that the invention is not limited in its application to the details of the particular arrangement shown, since the invention is capable of other embodiments. Also, the terminology used herein is for the purpose of description and not of limitation.

DETAILED DESCRIPTION OF THE DRAWINGS

- (99) Referring to FIG. **1**, the foot board **20** has a left board **21** and a right board **22**. Each board **21**, **22** has a central concave cutout so as to encircle the sea scooter **1** at about a midpoint of the longitudinal axis A of the sea scooter **1**. A latch **24** locks the left board **21** to the right board **22** around the sea scooter **1**. A left strap **25** attaches the left board **21** via a loop **27** to the hook **7**. A right strap **26** attaches the right board **22**.
- (100) Boots L and R are each attached to the board by an attachment structure. Such an attachment structure may comprise bindings similar to those used for a wakeboard, or water slalom skiing, or water skiing, or snowboarding, or those used for SCUBA fins, or quick dismount boots. A literal boot need not be used, as a user's bare foot may be secured by an attachment structure similar to that of a SCUBA fin, with the foot inserted into a recess or loop, and a loop secured around the heel to hold the foot in place. Where boots are used, the bindings may comprise Velcro straps, ski or snowboard-type bindings. Another embodiment is possible utilizing bindings for boots such as are used for mountain bike pedals, where a snap fitting snaps into place, but may be easily dislodged from the pedal by a deliberate motion of the user's foot. Further attachment structure are discussed below. It is advantageous for such attachment structure to allow for quick-disconnect, so that the rider may easily snap his or her foot out of the attachment structure. It is understood that as used herein, the control of the throttle of the device with the user's foot encompasses the concept of the user's foot being within a boot or the like.
- (101) Referring next to FIG. **2**, the foot board **200** attaches the same way as embodiment **20** but without the straps **25**, **26**. For all embodiments bungee cords or straps can be added for assisting with securing the foot board to a sea scooter.
- (102) Referring next to FIG. **3**, the handles **3** are received by suitable indents on the left board **310** and right board **320** of foot board **300**.
- (103) Referring next to FIG. **4**, a solid foot board **400** has a central hole to fit over the motor housing **2** above the handle **3**. The taper of the motor housing **2** helps sleeve the foot board **400** to the sea scooter **1**. During use, the propulsive force of sea scooter **1** will tend to keep it secure in the central hole of foot board **400**. The sea scooter **1** may be further secured and stabilized to the foot board **400** by the same means previously discussed above.

(104) Referring next to FIG. 5, a foot board **500** is formed with twin openings for receiving two sea scooters **1a** and **1b**. A left foot board section **510** has a concave opening that fits over the sea scooter motor housing **2b**, and a right foot board section **520** has a concave opening that fits over the sea scooter motor housing **2a**. The left board loop **502** has a bungee cord or strap **504** attached to handle **3** of sea scooter **1b**, as well as a loop **508** attached to opposite handle sea scooter **1b**. Likewise, right board loop **501** has a bungee cord or strap **503** attached to the outer handle of sea scooter **1a**, as well as a loop **505** attached to inner handle **3** of sea scooter **1a**. The left foot board section **510** may be separated from the right foot board section **520** by a detachable connector **502**, such as a latch between the two board sections. This allows the device to be disassembled for easier transport.

(105) Referring next to FIG. 6, a self-contained battery foot board **700** has a left board **701** and right board **702** integrated with the housing **706** of a water propulsion unit **705**, which may comprise a motorized electric propeller powered by lightweight Lithium batteries **703** and **704** sealed watertight within board **700**. Water enters into port **707** of the water propulsion unit **705**, and is discharged via a propeller from lower port **708**. FIG. 7 is an overhead view of the embodiment in FIG. 6. As will be discussed herein, in an embodiment of the device, the propulsion unit can be a trolling motor, as set forth herein, which typically consists of a main torpedo shaped body with a propeller.

(106) In FIG. 8, a different embodiment is shown in which foot board **800** is separable into left and right halves **801** and **802**, each with its own separate battery-powered propulsion unit **705a** and **705b**. As used herein, the term “half” does not literally require that the board be split evenly, and it should be understood that separating the board into two portions of unequal width is encompassed herein so long as the board is otherwise able to support a foot on each of the separate portions. As used herein, the term “portion” of a foot board may be used interchangeably with “half” or “halves” of the foot board.

(107) Here again, slim-profile Lithium ion batteries **703** and **704** are watertight sealed within the board, with sealed electrical leads extending out to the motors of the propulsion units. The user can lock the left to the right board using locking latch **803**, but in a preferred embodiment, latch **803** allows the left and right halves of board **800** to swivel with respect to one another, such that the user can tip one foot forward while rocking the other backwards, allowing for more versatile directional control when the device is in use. Such a latch might comprise an elastic connection—such as an elastic strap or spring—that allows the halves of board **800** to swivel, while also biasing them to return to a neutral position.

(108) A secure lateral connection between halves **801** and **802** can be aided by a male rod projecting outward along the central axis of the board **800** from one of the halves, wherein the rod is configured to mate into a hole on the corresponding side of the other half of the board, thereby allowing one half of board **800** to twist relative to the other half about an axis passing through the center of the rod.

(109) A throttle controller **850** for the propulsion units could be wireless or with a wire **851** as shown. A single controller **850** could be configured with separate throttle controls for the propulsion units **705a** and **705b**, or each propulsion unit could be paired with its own separate throttle controller. Usually, both units **705a** and **705b** would be controlled at the same speed, but allowing separate throttling will give the user more maneuverability. A microprocessor in the throttle controller could be configured to ensure that the thrust from one of the propulsion units always matches the other propulsion unit, or that the speed differential between one propulsion unit and the other never exceeds a certain threshold. Allowing separate throttle control for the two propulsion units also allows one to be placed into reverse thrust while the other provides forward thrust, thereby allowing the user to spin more quickly. And allowing the user to vary the relative thrust force of the two propulsion units will allow for greater control and maneuverability. FIG. 9 is a top plan view of the embodiment shown in FIG. 8.

(110) Referring next to FIG. 10, a foot board **900** is shown with individually pivotable feet as discussed with respect to the embodiment in FIG. 8. A linkage **901** is provided as a connector having a rotary bearing that enables rotation about an axis running through the board halves. Note that although the foot board has been shown in this and the preceding figures as having a flat surface, it is also possible to hydrodynamically shape the foot board surface to be curved to decrease water resistance when the device is in operation. For example, the edges of the foot board can be made to curve downward away from the boot mounts to allow water to more easily flow around them.

(111) Although the propulsion units depicted in FIGS. 6-10 have been shown as flat propeller units, it has been found that the device works very well with trolling motors used as the propulsion units. A trolling motor is an underwater electric propeller that is typically attached to a long rod and used as a makeshift outboard motor on small one-or two-man watercraft. A good trolling motor can generate 50 lbs or greater of thrust force, and there are models that are even substantially more powerful than that, supplying well over 100 lbs of force. Trolling motors are thus notably more powerful than prior art hand-held propulsion unit motor. As used herein, the term “trolling motor” is not limited literally to motors marketed as trolling motors, but to any electric propeller motors of similar construction or power. An example of a suitable trolling motor is a Haswing Protruar 24v, 2.0 hp motor, which is rated at 110 lbs of thrust; or a Minn Kota Saltwater Riptide, which is rated at 101 lbs of thrust; or a Newport Vessel, which is rated at 55 lbs of thrust.

(112) A commercially available trolling motor such as those just identified may need retrofitting for operation at depths greater than about 30 feet. High pressure gaskets are known in the art of, for example, sealed underwater video-camera equipment, that are more suitable for operation at significant depth than the gaskets found on ordinary commercial trolling motors available as of the time of this writing. Many of such gaskets are often made of polyurethane material or similar polymer. Water-tight sealing for deep diving can also be achieved by designing the motor casing to have multiple rows of gaskets at the sealing joints. The negative space within the motor casing chamber may also be filled with oil to prevent water intrusion during deep diving, with inlet and outlet valves for draining and replacing the oil. High-quality mineral oil is non-electrically conductive and will work for this application, though professional grade transformer oil (as is used in commercial electrical transformers) may be preferable.

(113) Referring next to FIG. 11 the prior art sea scooter **1** has a handle **3** and **300** with a scooter throttle button **12** on each **20** side. A throttle lever assembly **161** may be fastened to handle **300** with a second throttle assembly **161** fastened to handle **3**. This embodiment has a cable **162** within a sheath that is connected to hand controller **163** that has an activate trigger **164**. Trigger **164** pulls the head **166** of control cable **167** so as to tilt the lever **165** against the scooter throttle **12**.

(114) FIG. 12 shows a close-up of an example of a throttle lever assembly. When the cable **162** is pulled, it causes lever **165** to push down on throttle button **12**. FIGS. 13A, 13B, 13C, 13D, and 13E, show the throttle assembly **161** on its own from various angles. In FIG. 13D, the lever **165** is shown in dots in the neutral OFF position. The lever **165** hinges around hinge shaft **165a** which is mounted to back **191**. The back **191** has bolts **192** fastening it to the block **193**. Set screw **194** secures the hinge shaft **190**. As can be seen, cable **162** terminates in end **166**, and when cable **16** is pulled, end **166** in turn pulls down on lever **165**, which then presses down on the throttle trigger. FIG. 14 shows an exploded view of an example throttle lever assembly.

(115) Referring next to FIG. 15, a scooter board **2000** has a mounting hole **2001** to receive a sea scooter. Brackets **2002** secure hose clamps **2003** to lock the sea scooter in mounting hole **2001**. A protective sheath **2004** may be used. A right foot plate **2005** has a heel pivot mount **2006**, so it can be moved out O or in I by the toe T of the right boot R. A reverse hook up is optional where the toe is pivoted and the heel moves in and out, as will be shown in FIGS. 22 and 23. As the toe T moves in I, the cable end **166** pulls the control cable **167**, and the lever **165** on the trigger assembly **161** is depressed into the scooter trigger. Thus, this embodiment enables the user to control throttle by

rotating their feet on the surface of the foot board, with the sprint returns tending to bias the feet back to a neutral position.

(116) FIG. **16** is a bottom perspective view of the embodiment shown in FIG. **15**, and FIG. **17** is a top plan view of the same embodiment. FIGS. **18** and **19** are like FIGS. **16** and **17** except with a reverse mounting of the control cables **167**. As can be seen, the spring ball **2010** pushes the flat spring **2011** inward during acceleration. As can be seen, the spring **2011** returns the lever **165** to neutral when the user stops pushing in I.

(117) Referring next to FIGS. **20** and **21**, the boots L and R are mounted to their respective foot plates **2030** and **2005** by an attachment structure (as previously described in connection with to FIG. **1**). A hole **2300** allows the cable **162** to exit from under the respective foot plates. FIG. **22** shows a top perspective view of the device wherein the swivels **2002b** and **2002d** are located at the heel. FIG. **23** shows a view of the underside of the device from FIG. **22**. FIG. **24** is an exploded view of the device, and FIG. **25** shows the device with a sea scooter inserted.

(118) FIGS. **26**, **27** and **28** demonstrate how a sea scooter rigged with a wired or wireless throttle controller may be mounded to an L-bracket **3003** attached to a body plate **3001** or **3004** with which has shoulder straps **3002** for a swimmer. Straps **2003** secure the sea scooter to the L bracket **3003**. This L-bracket configuration provides a versatile mounting means. FIG. **30** shows a foot board embodiment **5001** that uses L-brackets **3003A** and **3003B** and straps **2003** to secure a left and right foot board with boots L and R.

(119) Referring next to FIGS. **31**, quick disconnect boots RQ and LQ have a bottom flange **3100** that fits into the groove **3101** on respective left and right foot boards **3102** and **3103**. When the sliding lever arm **3999** is in the neutral position NU, the flange **3100** can be inserted into the groove **3101**. When the lever arm **3999** is moved to the lock position LK shown in dots and the movement for which is shown by arrow LK, the rod **3109** has passed through a hole HL in flange **3100**, locking the boots onto respective boards **3102** and **3103**. FIG. **32** shows the arms in the locked position. The boots may be released by pulling the arm **3999** back to the neutral position.

(120) Referring next to FIG. **33** an electronic foot control board **3300** is shown-a plan view of the underside (FIG. **34** shows the device from the top side). A base **3301** has a forward carry handle **3302**. A propeller motor **3303** may be a DC voltage waterproof type powered by a rechargeable Lithium ion battery. Power leads and wiring are water tight and may be sealed in silicone or the like. A left foot pedal **3305** has a swivel mount **3306** to the base **3301** (a corresponding swivel mount in the right foot board is shown but not labeled). The user's boots strap or interlock securely to the swivel pedal via an attachment mechanism (as previously described in connection with to FIG. **1**), and the swivel pedal is then capable has a hole that receives and locks to a projection from the underside of the toe of the user's boot, allowing the user to twist their feet in the base **3301** about an axis running through their toes, causing the heel ends of their boots to move side to side at the rear end of the base **3301**. Note that this configuration could be easily reversed so that the heel end of the boots mounted to a swivel, and the toe end of the boots was allowed to move side to side.

(121) A magnet (or equivalent transmitter) **3308** is attached to a rear section of the foot pedal **3305**, and a magnet (or transmitter) sensor **3307** is connected to the base **3301**. The sensor **3307** has an electronic connection to the motor speed controller **3309**. The motor speed controller may be a pulse width modulated (PWM) type. The sensor **3308** may be a hall effect type. The position of the magnet and sensor could be reversed by design choice. The motor speed controller **3309** is a software flow processor that reads the state of the magnetic sensor **3307** in the main loop. If the sensor **3307** has been activated, the processor **3309** checks if the motor is running. If the motor **3303** is running and the sensor **3307** is held in an activated state for greater than X seconds, motor **3303** is turned off. If the motor is running and the sensor is activated for less than X seconds, the speed is increased one increment (unless already at top speed, in which case nothing happens). If the sensor **3307** is activated twice in a row and motor is running, speed is decreased one increment

(unless already at maximum speed in which case nothing happens). If the motor is off, and the switch is held in activated state for greater than X seconds, motor is turned on at lowest speed.

(122) As a more general matter, it may be appreciated that by virtue of the swivel pedal mounts and sensors, the user is able to control the throttle of the propulsion unit by twisting their boot (and thereby the foot pedal) on the surface of the base **3301** about the axis of the swivel mount, with a sensor detecting the extent of movement of the opposite (moving) end of the boot, and translating the extent of that movement into a desired amount of throttle. A foot movement other than a swivel may be enabled to control throttle by, for example, including a spring-mounted pedal below the user's toes which functions in a manner similar to an ordinary automobile gas pedal. Such an embodiment is shown in FIG. **46**.

(123) In the alternative to using the degree of movement of the foot to control throttle, the sensor **3307** may comprise an electrical switch connected to an electrical circuit and a microprocessor. In the switch embodiment, the microprocessor may be programmed such that each tripping of the switch by a foot movement causes the propulsion unit to cycle through different levels of thrust. For example, each new trip of the switch can increase throttle until a last click drops the throttle back to zero. The processor might also be programmed to change thrust based on a particular pattern of tripping of the switch, such as increasing throttle based on two switch trips in rapid succession. Referring to FIG. **36**, and embodiment of a foot board **3601** is shown having propulsion unit **3611** and a foot pedal mounted to swivel **3606** and connected to spring return **3503** which tends to bring the foot pedal back to neutral position when the user does not exert any twisting force on the pedal. A switch **3617** with a button is affixed to a side extension of foot board **3601** and positioned such that it may be struck by the foot pedal when the user twists their foot and causes the foot pedal to pivot about swivel **3606**.

(124) Referring next to FIG. **34**, the propulsion unit **3309** has a propeller P shown in FIG. **35** below the base **3301**. As shown here, this propulsion unit is similar to that of a trolling motor (previously described) which provides more thrust than a conventional sea scooter. This design does not require any electronics to be mounted to the foot pedal **3305**. Only the magnet **3307** (shown in FIG. **35**) needs to be mounted on the swiveling foot pedal **3305**. A forward slot **3310** can guide the foot pedal **3305** with a stopper **3311** functioning as a guide post and a maximum travel stopper. A watertight power line supply tube **3325** is shown leading from the battery compartment within the board to the propulsion unit **3309**.

(125) Referring next to FIG. **35** a bracket **3501** secures the motor **3303** to the base **3301**. A right foot pedal **3502** and duplicate controls are optional. A kill switch **3508** has a tether **3509** to the leg of the user (not shown) wherein if the user becomes separates from the board, the user's leg will pull the tether and release the kill switch, turning the propulsion unit off. A spring return **3503** returns the foot pedal **3305** to a neutral straight ahead position. A platform spacer **3504** secures one or more batteries **3304**. Screws **3505** are shown as needed. A battery cover **3506** has fasteners **3507** to quick connect to platform spacer **3504**. A gasket traverses the top edge of cover **3506** and acts to seal the battery compartment when pressed against the spacer **3504**, and the spacer **3504** in turn has a perimeter gasket that engages with the underside of board base **3301**.

(126) An advantage of a board design such as that shown in FIG. **35** is that the board is formed and configured as having a thin profile of, for example four inches or less, and the use of flattened batteries allows the thin profile to be maintained. A thin board of this kind is easily carried by the user, and its total weight with the integrated flattened batteries might only be approximately 30-40 pounds when the balance of the board is constructed largely of lightweight polymer materials. As used herein, the term "integrated" refers not only to placement within the body of the footboard, but also encompasses direct attachment to or on the foot board.

(127) Referring next to FIG. **37** optional repair openings **3700** for the spring return **3503** are shown. Referring next to FIG. **38** the subsystem microcontroller **3309** C is programmed as shown in FIG. **39** or with many equivalent logic steps as known to one skilled in the art. A foot pedal

movement or a switch (not shown) starts **3900**. The logic in microcontroller **3309C**. The sensor **3308** is read at **3901**. If the sensor is activated in **3902** the logic proceeds to determining if the motor is running at **3903**. If the sensor held ON at **3904**, then stop the motor if the motor is running at **3905**. If the motor was OFF, then start the motor at **3906**. A double hit at **3907** either maximizes the speed at **3908**, or if already at maximum speed, it decreases the speed at **3909**, a single hit at **3910** can increase the speed one increment at **3911**. Other variants on this programming and function are possible. The purpose is to enable the user to control throttle by use of a motion of their feet on the foot board.

(128) Another computer-controlled system that is advantageous to employ with the disclosed devices is that of a depth-activated speed-limiter. In this embodiment, a depth gauge could be incorporated with the foot board, and electrically connected with the throttle control. Pre-set parameters could then be used to regulate the user's throttle based on depth, or the user could modify the parameters while the foot board is in use. Another kind of speed-limiter may be employed to pre-set the maximum speed of the foot board based on the level of skill of the user, or the anticipated diving conditions. Thus, the maximum speed of a beginner could be set lower, or the maximum speed could also be set lower for wreck-driving in close quarters.

(129) Referring next to FIG. **40** an alternate embodiment remote **4000** could either replace a foot pedal or augment a foot pedal embodiment for a backup or user choice. An antenna (not shown) would be needed on a microcontroller and receiver (that usually reaches with a radio frequency up to nine feet underwater). A speed up **4001** or speed down **4002** and stop **4004** button, and start button **4003** is shown. Such a remote **4000** could be attached like a watch to the user's wrist.

(130) Although the present invention has been described with reference to the disclosed embodiments, numerous modifications and variations can be made and still the result will come within the scope of the invention. No limitation with respect to the specific embodiments disclosed herein is intended or should be inferred. Each apparatus embodiment described herein has numerous equivalents.

(131) Referring now to FIGS. **41A** and **41B**, an embodiment is shown in which the foot board **4100** is separated into left and right halves **4105A** and **4105B** that are releasably connected by magnetic surfaces **4107A** and **4107B** that form a magnetic linkage when connected. Surface features of the boards, such as swiveling foot pedal mounts and throttle control, are not shown for simplicity. Lithium ion batteries may be sealed within the bodies of the left and right boards, with sealed leads connected to the propulsion units **4111A** and **4111B**, shown here as trolling motors. As shown in FIG. **41B**, the two halves of the foot board may be snapped together by magnetic attraction. However, the strength of the magnets may be set so as to allow the user to unsnap the two board halves by applying a deliberate spreading force, or by sliding the halves parallel past each other. The magnets may also be configured so as to allow the two foot board halves to pivot individually from each other while remaining connected. Of course, two foot board halves may be joined together by rigid latches, or by a male-female rod connector to form a single connected board, but such a single connected board would not enable relative movement of one half to the other.

(132) Referring next to FIGS. **42A**, **42B**, and **42C**, a foot board **4200** is shown split into halves **4205A** and **4205B**. Surface features of the boards, such as swiveling foot pedal mounts and throttle control, are not shown for simplicity. Lithium ion batteries may be sealed within the bodies of the left and right boards, with sealed leads connected to the propulsion units **4211A** and **4211B**, shown here as trolling motors. A linkage **4210** holds the halves **4205A** and **4205B** together. This linkage **4210** may comprise a rigid rod of fixed length, mounted by bearings or swivel mounts in the inner sides of each half **4205A** and **4205B** to allow the halves to pivot with respect to one another. For example, one half of the board may protrude a male rod that mates with a bearing on the opposing half of the board. Alternatively, linkage **4210** may comprise a flexible connector such as a heavy polymer material that tends to return to a straight rod shape, but which may be bent or twisted in infinite directions under force by the user's boots, as shown in FIGS. **42B** and **42C**, thus allowing

the halves **4205A** and **4205B** to assume a wide range of different relative positions and orientations with respect to one other. Alternatively, the linkage **4210** could be made of a limp yet durable material (such as polymer rope) that allow completely unconstrained relative movement of the halves **4205A** and **4205B**, while preventing the halves from separating more than the pre-determined distance of the linkage. As known in the art generally of straps, such linkage can be made length adjustable.

(133) Referring to FIG. **43**, an embodiment is shown of foot board **4301** wherein a string of watertight LED lights **4311C** encircles the perimeter of the board, and may be used to locate divers underwater in dark or murky conditions. Further strings of LEDs **4311A** and **4311B** are shown encircling the rim on enlarged battery casings **4303A** and **4303B** designed to accommodate large sized batteries for greater battery life for the combined motor and lighting system.

(134) Referring to FIG. **44**, an embodiment is shown of board **4401** that is provided with optional dive weights **4404** that may be inserted into correspondingly shaped slots in board **4401**. The board may be constructed so as to be neutrally buoyant in fresh water, with the ability to add weights as ballast in salt water.

(135) Referring to FIG. **45**, an embodiment is shown of foot board **4501** that includes a small pressurized air tank **4503** filled with compressed CO₂ or the like capable of being released by the user to inflate bladder **4505**, which can be used to automatically send the board **4501** to the surface of the water if the user becomes separated from the board or otherwise wants to send it to the surface separately. A release valve **4507** is also provided.

(136) Referring to FIG. **46**, an embodiment **3300A** of the foot board **3300** previously shown in FIG. **34** is presented wherein the throttle switches are toe pedals **4602**.

Accessories and Alternative Configurations

(137) Because of the newness of the product designs herein introduced, it is desirable to have a series of accessories and alternative configurations to assist users in customizing their products. Several such useful and novel accessories and configurations are shown in FIGS. **47A** thru **70**.

(138) FIG. **47A** depicts an embodiment of the device **4750** wherein a motor-propeller combination **4773**—such as a trolling motor—is encased within a cowling **4775**, shown here as a cylindrical cowling attached to the foot board **4706**. This kind of propulsion may be referred to as ducted fan propulsion. The cowling **4775** serves the purposes of protecting the motor and its propeller blades from harmful impacts, protecting the user and other persons or animals in the water from being struck by the spinning blades, and focusing the thrust from the propellers in a more defined direction. Furthermore, if the cowlings are extended beyond the length of the motor (as in FIGS. **48A** and **48B**), they can also be used as a stable base to stand the board up on a flat surface without that surface contacting—and potentially damaging—the motors. Also shown is internal batteries **4777** and boot holsters **4771**.

(139) FIG. **47B** depicts an embodiment of the device **4700** wherein a motor-propeller combination **4703**—such as a trolling motor—is encased within a cowling **4705**, shown here as a cylindrical cowling attached to the foot board **4706**. This kind of propulsion may be referred to as jet propulsion. The cowling **4705** serves the purposes of protecting the motor and its propeller blades from harmful impacts, protecting the user and other persons or animals in the water from being struck by the spinning blades, and focusing the thrust from the propellers in a more defined direction. Gratings **4715**—such as a grid of thin rigid plastic or wire—on the bottom openings of the cowlings help prevent damage to the propellers and injury from someone coming into contact with them. Furthermore, if the cowlings are extended beyond the length of the motor (as in FIGS. **48A** and **48B**), they can also be used as a stable base to stand the board up on a flat surface without that surface contacting—and potentially damaging—the motors. Also shown in FIGS. **47B** are internal batteries **4707** and boot holsters **4701**.

(140) FIGS. **48A** and **48B** show an alternative embodiment of the device **4800** having a cowling **4803** that can be used to concentrate the flow of water from the propellers. As used herein, this

form of propulsion may be called jet propulsion. A stator fin within the cowling can help direct and stabilize the flow of the water, saving motor energy. A pre-swirl stator may also be used in front of the propellers to stabilize the flow of water being pulled through before it hits the propellers, leading to more engine efficiency and power savings.

(141) FIGS. **49A-49C** depict a motor mounting bracket **4901** that can attach to a person leg calf, leg thigh, waist, back, chest or arms. The bracket itself can be attached to the user with straps **4909** and **4911**, and can have a soft molded contact mount for contacting the user's skin or clothing (not shown). The molded mount may be part of a prosthetic that wraps around the user's leg that incorporates elastic or straps to hold the prosthetic securely in place, and thereby support the bracket and the weight of the attached motor. The bracket **4901** may be incorporated as rigid channel within such prosthetic. The bracket **4901** also allows the motor **4915** to be slid up and down its length, and locked in place at a desired position, as well as at angles relative to the bracket. This can be accomplished, for example, using a U-channel type bracket with pin holes **5913** for a mating spring button. An example is shown in FIG. **60**. FIG. **65** further depicts a spring-loaded handle **6501** that can be used to move and fix the position of the motor on the bracket. As swivel mount can also allow the motor to be angled relative to the brackets.

(142) Electricity can be brought to the motor via waterproof wires (not shown) embedded in the channel and with sufficient slack to allow the motor to be moved along the length of the channel. In this embodiment, the battery pack can be located other than in a board at the user's feet, such as in a back-mounted pack with electrical cables that run to the motors. A foot pad **5201** and strap at the base of the channel (also shown in FIGS. **52** and **61**) can be used to help secure the bracket to the side of a user's leg and maintain the bracket's position relative to the leg, while a molded mount and/or prosthetic between the bracket **4901** and the side of the user's leg (not shown) can also be used to ensure that the bracket **4901** stays in a fixed position relative to the user's leg.

(143) The foot plates **5201** may be locked in an open position using, for example, the lock depicted in FIGS. **63** and **64**. The lock comprises a first plate **6303** with a channel on the underside of one portion of the foot plate **5201** opposite a second plate **6301** with a channel on the underside of the opposite portion of foot plate **5201**, which is separated from the first portion by a hinge. When the foot plate **5201** is deployed, the lock member **6305** may be slid from the channel on the second plate **6301** partly into the channel of first plate **6303**, thereby locking the foot plate in a deployed position.

(144) The straps **4909** and **4911** may employ buckles or Velcro, or any other convenient strap binding mechanism. The brackets **4901** can also allow the motors **4915** to be slid off the brackets **4901** entirely so that they can be stored when not in use, or swapped out for different motors. A detachable waterproof linkage (not shown) is employed to couple and uncouple the motors from the wiring that supplies their power from the battery.

(145) As shown in FIG. **51**, a rotating pivot joint can be used to mate the motor **5101** to the bracket **4901** so as to enable the motor **5101** to rotate relative to the bracket **4901**. The rotating pivot joint can be locked in position by the user when the desired position is reached.

(146) As shown in FIGS. **55C**, battery packs **5503** can also be located on the user's boots **5501**, with waterproof electrical wires connecting the batteries to the bracket-mounted motors. In an embodiment, the wires may be contained in a self-coiling mechanical spool that allows slack to be let out when it is desired to slide the motors up the bracket, such as upon ingress or egress to the water. The self-coiling spool will also automatically retract the wires within the spool compartment when the motors are moved closer to the batteries. The electrical wires can also be coiled so as to not dangle excessively when slack is taken back in. However, care should be taken that the wires are not allowed to have too much slack such that they could become tangled with the user's gear, equipment or underwater obstacles. Containing the wire within the channel of the bracket **4901** will aid in this, such that wire slack is contained within the bracket channel. In alternative embodiments, the batteries may be placed elsewhere on the user-such as in a backpack, or on a waist-belt, or on

the bracket itself-and similar techniques used to manage the wiring.

(147) As shown in FIGS. **55A** and **55B**, the two foot plates may be locked together with a bar **5313**, connected at each foot pad by a mating to a female lock **5317** with spring bearings. Locking the foot plates together will help stabilize the user's legs underwater when the device is in use. The locking bar **5313** can telescope and lock in position using a spring button capable of going into any of a multiplicity of holes along the bar's length, as shown in FIGS. **66** and **67**. The bar **5313** can also have a rotary—or ball and socket—joint along its length to allow the motors foot pedals to orient back and forth relative to each other. The bar **5313** can be designed to be easily released with one grabbing motion of the user's hand and arm, thus allowing the user to move their feet separate to walk.

(148) In one mode of use, a person would use the device described herein in the ocean, a river, or a lake, and upon wanting to exit the water, would come up to the beach/banks, and by unlocking the bar between the foot pads, and also lifting the motors up along the brackets and locking them in place, the user could walk up out of the water onto the shore. The foot pads may also fold up toward the leg and be secured there so as to be out of the way. In general, the device can be made to fold up onto the user's legs to also for ease of ingress and egress from the water. FIG. **50** depicts a motor toting system whereby a motor may be attached to a user's utility belt via a carabiner **5001** for ease of carry.

(149) As shown in FIGS. **55D** and **55E**, the bar **5507** between the foot pads can have a hinge, such that the bar can simply be retracted up along the user's leg, without having to remove the bar entirely. An attachment along the user's leg (whether connected to bracket or strapped onto the leg) can be used to connect to the free end **5511** of the retracted bar **5507** and hold it in place, such as during ingress and egress from the water. A female holster **5513** holds the free end **5511** during operation of the device.

(150) FIGS. **53A** and **53B** show a carrying strap system whereby a carrying strap **5301** is attached at two points to the board **5302** using male and female hooks **5305** and **5303**. Carabiners or the like may also be used. The strap's length can be varied to allow carrying by hand, or throwing it over a shoulder.

(151) FIGS. **54A** and **54B** depict a custom carrying bag enclosure **5401** for the board **5302**, whereby the bag **5401** is shaped to fit around the board **5302** and boots and zipper or seal up with the motors exposed. Straps can be attached to the bag to allow carrying by hand or like a backpack.

(152) FIGS. **56A** and **56B** show how a small electricity-generating turbine **5603** can be attached to the device (shown here as attached to a cowling **5601**, but it could be attached elsewhere as well). The blades of the turbine **5603** are oriented like propeller blades such that forward movement through the water will turn them. The electricity generated by the turbine can be used for any purpose, including powering of underwater accessory devices like cameras, or it can be fed back into the main device battery. A switch (not shown) can be employed to toggle between charging an external device, and running back to the device's own battery, such that power is not wasted if the turbine is otherwise is use during a given underwater activity.

(153) FIGS. **57A** and **57B** show how the motor may be mounted to a prosthesis. Wiring (not shown) runs to a waterproof battery pack attached elsewhere to the user, such as in a backpack. The other leg may use a mounting bracket as otherwise described herein.

(154) FIG. **58** depicts the attachment of inflatable bags **5801** to the device (shown here as attached to the back of the boots) to increase buoyancy. These may be manually inflated by the user, or may come equipped with an attached compressed air canister with a release valve that the user can rapidly pull/activate if buoyancy is needed immediately in an emergency.

(155) FIG. **59A** and **59B** show a backpack-style set of straps **5902** that can attach the motors via male and female mounting hooks **5905** and **5907**, carabiners, or the like.

(156) FIG. **62** depicts an electronic readout **6201** (optionally employing LED lights) on the device between the user's legs and visible to the user by looking down towards their feet. The readout

6201 displays information to the user, such as depth, relative water velocity, and electric charge remaining in the battery. A large read-out in this location will be easily seen by the user during use of the device.

(157) FIG. **68** depicts a direct attachment between the motors and the boots. Here, the battery has been made integral with the shoe.

(158) FIGS. **69A** and **69B** depict a chargeable battery in a backpack that can be used to re-power the battery of the board. This enables the user to carry a back-up power source with them when otherwise away from a charge port for an extended period of time.

(159) FIG. **70** depicts a direct wired battery charging embodiment that allows charging via a standard wall socket with a power converter. A re-sealable watertight connection (not shown) accepts the plug on the device.

(160) FIGS. **71-75** depict an embodiment **7100** of the device comprising two rigid cylindrical calf sleeves **7101** for fitting around the right and left calves of a wearer. The term “rigid” here encompasses the kind of rigidity associated with hard plastic or metal, as well as the kind of rigidity associated with plastic or rubber-like materials that are mostly stiff but capable of slight deformation by a human hand, and in any event sufficiently rigid to securely mount the motors and other equipment disclosed herein without meaningful deformation during underwater operation. The sleeves **7101** are shown in FIGS. **71-75** as having solid surfaces, but in alternative embodiments could have openings, so long as the sleeves are sufficiently rigid to securely mount the motors and other equipment disclosed herein without meaningful deformation during underwater operation.

(161) Each sleeve **7101** is composed of two halves split along the long axis of the sleeves **7101** such that the halves can open to allow a wearer to place a lower leg into each of the sleeves **7101** and then re-close the sleeves **7101** with their lower legs inside. Straps **7103** connect to buckles **7105** to allow the user to tighten the sleeves **7101** about their lower legs.

(162) An inflatable form fitting calf bladder **7113** is disposed within each sleeve **7101**, and may be pumped full of air by using the pump button **7115** on each sleeve **7101**. The pump button may be designed such that twisting it will act to deflate the bladders **7113**. The user can inflate the bladders **7113** to provide a better fit between their lower legs and the inner walls of the sleeves **7101**, as well as to provide cushioning.

(163) Each sleeve **7101** has a waterproof battery pack **7111** disposed on its dorsal side with a removable battery cartridge. A power line runs internally through the device to adjacent propulsion units each comprising a motor **7107** and a propeller protected by a guard **7109**. The propulsion units are mounted to the outer sides of the user's legs, as shown in the drawings, though might alternatively be mounted on the dorsal (back) or ventral (front) side of the user's legs by positioning the battery packs in a different location about the sleeves. In alternative embodiments, the batteries (or single battery) may be worn elsewhere on the user, such as in a belt pack or backpack, and connect to the propulsion units by watertight power chords.

(164) The two sleeves **7101** can be connected to each other by a rigid telescoping bar **7131** with halves **7131a** and **7131b**. The rotating tightening clamp **7129** may be twisted by a user's hand to allow the telescoping bar halves to be fixed in position relative to each other, or rotated to unclamp and allow the halves to telescope relative to each other to allow the bar **7131** to widen or shorten. The free end **7134** of bar **7131** snaps and locks into a holster **7125**, but can be released by pressing release button **7126**. This then allows the bar **7131** to fold upward about pivot pin **7143** (shown in FIG. **73**) into storage recess **7121** as shown in FIG. **75**. At some point along either the length of the bar **7131**, or at one of the joints that attaches the bar **7131** to either sleeve, a rotary joint can be employed so as to allow the user to twist one sleeve relative to the other about the axis of the bar **7131**. The positioning of this rotary joint will be referred to as along the length of the bar (even if at the joint with the sleeve) for simplicity.

(165) The bar **7131** may also be mounted higher up on the sleeves rather than near the ankles. Also,

the connection joints between the bar and the sleeves can be rotary joints with multiple degrees of freedom (including ball and socket joints) such that the user can maneuver one sleeve relative to the other in many directions, which the bar keeps the sleeves at a fixed distance apart from each other. In other embodiments, the bar may allow telescoping while in use to allow the user to widen or shorten the length of the bar while using the device underwater. A tensioner clamp on the bar can be used to control the resistance to make it harder or easier for the bar to telescope.

(166) A ball and socket joint may also be positioned midway across the bar **7131**, and designed so as to allow the user to maneuver one sleeve relative to the other in a wide range of directions and differing orientations, while also preventing the two halves of the bar on opposite sides of the ball and socket joint from bending any more than a pre-determined angle relative to one another, for example, no more than 45 degrees, thereby maintaining a desired minimum distance between the two sleeves. In the foregoing embodiment, the telescoping element(s) of the bar **7131** may be placed on one or both halves of the bar **7131** to the side of the ball and socket joint.

(167) In all embodiments employing a bar **7131**, the configuration of the bar **7131** and the various joints within it or that attach to the sleeves can be configured so that there is a minimum fixed distance maintained between the two sleeves, such that the bar **7131**-even if capable of bending or twisting or telescoping-is rendered mechanically incapable of bending or twisting or telescoping beyond a certain limit that would otherwise allow the two sleeves to come closer than the minimum predetermined distance. This can be accomplished by designing the joints to be incapable of moving past a certain extent and/or designing the telescoping aspect to have a limited range. Thus, while a user might be able to bend, or twist, or telescope the bar during operation of the device such that the sleeves exceed the minimum predetermined distance from each other, the bar will not allow the sleeves to come within less than the predetermined distance from each other.

(168) While the bar **7131** in FIGS. **71-75** is shown as fixed to one of the sleeves, the bar can also be designed so as to be fully removable from both sleeves. Whether fixed to one of the sleeves or fully removable from both sleeves, the bar will be referred to herein as “removable,” meaning that the user can manipulate it (or its connection joints) by hand to decouple (or re-couple) one sleeve from the other.

(169) The foot stirrup **7135** accommodates one of the user's feet, and is used as a throttle to control the speed of the motors **7107**. In the embodiment shown, the throttle is controlled by the user turning their foot left or right, with the foot pointed straight ahead being no throttle, and throttle increasing as the user turns their foot outward in the stirrup **7135**, causing the foot stirrup **7135** to rotate outward about an axis running through the center of the length of the sleeve. An equivalent system could be designed to react to the user angling their foot up or down, like on a car gas pedal. A toothed knob **7141** can be turned by the user to adjust the resistance level of the stirrup to twisting, so as to make it harder or easier for the user to turn the stirrup with their foot to change throttle.

(170) The electronic throttle control system within the device (not shown) adjacent to the stirrup can be connected to the electric motor **7107** on the opposite leg by several alternative means. First, a connection may be made through the bar **7131** whereby a waterproof sensor contact is made between the free end **7134** of the bar **7131** and the holster **7125**. This sensor contact can comprise a direct conductive electric connection, or alternatively a laser light passing through sealed transparent windows between the free end **7134** and the holster **7125**. In another embodiment, a wired electric cable (not shown) can connect the throttle control system to the motor on the opposite leg. In another alternative embodiment, a wireless transmitter connected to the throttle control system can wirelessly send a signal to a wireless receiver attached to the opposing motor. In another embodiment, each sleeve **7101** can have its own foot stirrup **7135** and throttle control system such that the user can separately control the throttle of each motor using both feet.

(171) The user can rotate the motors **7107** within their mounts by twisting the clamp screw **7149** to loosen it, then rotating the motor, then twisting the clamp screw **7149** again to tighten it and lock

the motor in place. Alternatively, the motor can be mounted on overlapping cylindrical bars with holes along their circumference that allow a set pin to be inserted through them to lock them relative to each other, and then the set pin removed to allow them to rotate relative to each other. (172) Each sleeve **7101** has a toothed track **7145** that allows the motor to be slid up and down the side of the sleeve. A pinchable set clamp **7147** locks the motor in position relative to the track **7145**, and can be squeezed by the user to allow it to slide along the track **7145**.

(173) A key slot **7138** is capable of receiving a key **7136** for activating the device. The key **7136** can contain a magnetic strip with a personalized access code, or other known means of key card access.

(174) FIG. **76** shows a circuit diagram for an embodiment of the device. The operations are controlled via the processors on the system control board **7603**. Hall effect switch **7601** receives input from the throttle that ultimately dictate the amount of power flowing from the batteries **7611** and **7614** to the electric motors **7604** and **7605**. Hall effect switch **7602** receives input from the key to control the on/off/kill functions. Relays **7606** and **7607** control signaling. The system is also equipped with circuit breakers **7608** and **7609**. Battery charge indicator **7610** is controlled by the system control board. Battery charge indicators **7612** and **7613** display charge information for each of the batteries **7611** and **7614**. Other devices can be wired into the system control board, such as a depth/pressure sensor that kills power or reduces throttle automatically based on depth, a depth display readout, or a trip sensor for inflating an emergency bladder upon exceeding a certain depth.

(175) A variety of simple programming in the device's on-board computers and circuit boards are advantageous in increasing safety. For example, a low battery alarm can warn the user to return to a safe location before the battery dies. The alarm can be auditory, or could take the form of a distinctive vibration. An emergency back-up power may be activated by the user affirmatively activating a switch, such that there is always some reserve of power and the user cannot inadvertently run out. The motors may be programmed to start pulsating if their battery is running low.

(176) A sensor in the boot can also detect when they are locked in place, such that electricity will not flow to the motors unless the boots are fully locked in.

(177) A speed control can also be employed to automatically slow the motors to a certain pre-determined speed when they are operating a particular depth, or at a particular distance from the bottom (requiring some short-range impact detection emitter), or at a particular location (requiring on-board GPS), such as a protected marine environment.

(178) Although the invention has been described in terms of exemplary embodiments, it is not limited thereto. Rather, the appended claims should be construed broadly, to include other variants and embodiments of the invention, which may be made by those skilled in the art without departing from the Scope and range of equivalents of the invention.

Claims

1. A device, comprising: an assembly for mounting to a user's foot, wherein the assembly comprises a foot pedal, a battery sealed in a watertight compartment, and a battery-powered underwater propulsion unit; a throttle control system integrated with said foot pedal, wherein said foot pedal is capable of a swivel motion about an axis parallel to said user's lower leg in response to a swivel motion of said user's foot about said axis, said motion controlling the throttle of said propulsion unit, wherein said propulsion unit comprises a motor-propeller combination within a cowling.
2. The device of claim 1, wherein said motor-propeller combination and said cowling are configured for ducted fan propulsion.
3. The device of claim 1, wherein said motor-propeller combination and said cowling are configured for jet propulsion.

4. The device of claim 1, further comprising a pre-swirl stator mounted at the intake of said motor-propeller combination.
 5. The device of claim 1, wherein said cowling has a base configured to rest flat on a surface to allow said device to stand on a surface without damaging the motor-propeller combination.
 6. A device, comprising: an assembly for mounting to a user's foot, the assembly comprises a foot pedal, a battery sealed in a watertight compartment, and a battery-powered underwater propulsion unit; a throttle control system integrated with said foot pedal, wherein said foot pedal is capable of moving up or down in response to the motion of said user's foot up or down, said motion controlling the throttle of said propulsion unit, wherein said propulsion unit comprises a motor-propeller combination within a cowling.
 7. The device of claim 6, wherein said motor-propeller combination and said cowling are configured for ducted fan propulsion.
 8. The device of claim 6, wherein said motor-propeller combination and said cowling are configured for jet propulsion.
 9. The device of claim 6, further comprising a pre-swirl stator mounted at the intake of said motor-propeller combination.
 10. The device of claim 6, wherein said cowling has a base configured to rest flat on a surface, allowing said device to stand on a surface without damaging the motor-propeller combination.
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